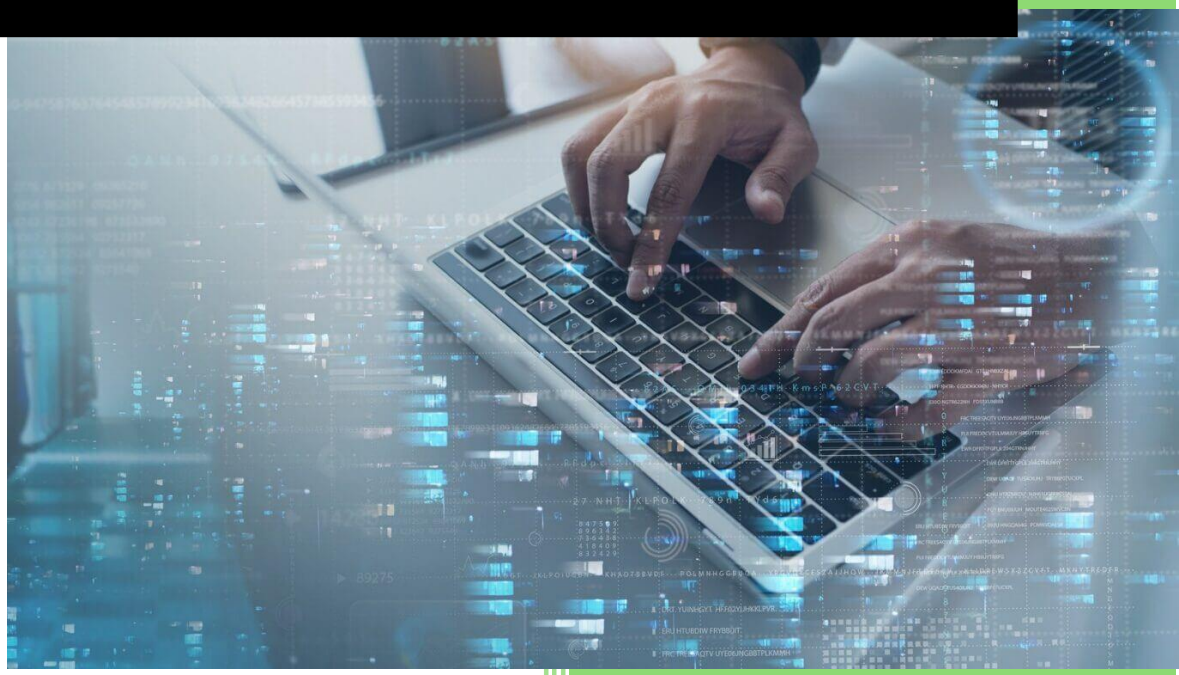
2025

HUMAN COMPUTER INTERACTION



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ST10447425

11/14/2025

StudySync – Part 3 Prototype Report

1. Introduction

This Part 3 submission presents the final high-fidelity, interactive prototype of **StudySync**, a web-based student collaboration platform. StudySync is designed to enhance academic engagement by enabling students to organise assignments, motivate peers, share updates, and interact through reactions, comments, and celebration tools. The development process applied established Human–Computer Interaction (HCI) principles to ensure usability and user satisfaction (Nielsen, 1994; Shneiderman & Plaisant, 2010).

2. Goals of Part 3

Part 3 addresses the implementation and refinement of the StudySync prototype. The primary objectives are to convert Part 2 designs into a functional coded system, enhance ease of use, and ensure accessibility. These objectives align with recommended HCI lifecycle processes, which emphasise iterative development and improved user experience (Shneiderman & Plaisant, 2010).

3. Improvements from Part 2 Comments

After the input from Part 2, a number of changes were made:

- Better layout spacing and visual hierarchy.
- Toggle buttons for easier navigation.
- Enhanced customization via student profiles.
- The incorporation of interactive multimedia features, like live video previews.

These improvements are in keeping with usability standards that prioritize user control, consistency, and visibility (Nielsen, 1994).

4. Modified User Needs

Functional Conditions

- The ability to sign up and log in.
- Creating, modifying, and monitoring tasks.
- Peer interactions include like, commenting, and celebrating.
- Group code generation and tool sharing.
- Uploading avatars and modifying profiles.
- The ability to preview videos live.

Non-Functional Conditions

- ❖ An easy-to-use interface that complies with usability guidelines (Nielsen, 1994).
- ❖ Consistent layout, iconography, and color palette.
- ❖ Adaptability to different screen sizes.
- ❖ WCAG guidelines are adhered to by accessibility features (W3C, 2018).

5. Part 3 System Architecture

- JavaScript is used for logic and interaction
- HTML5 is used for page structure
- CSS3 is used for styling and appearance
- and LocalStorage is used for temporary data persistence.

According to UI development guidelines, this structure facilitates lightweight, responsive, and user-centered design (Shneiderman & Plaisant, 2010).

6. Description of the High-Fidelity Prototype

- ★ A revamped dashboard for peer motivation and assignment organization is part of the final StudySync prototype.
- ★ Features for authentication (signup, login, logout).
- ★ Student profiles are customizable.
- ★ An interactive feed for peer interaction.
- ★ Animation-based responses and notification components.
- ★ A prototype for a live video preview that makes use of the camera on the device.

According to interaction principles, the design prioritizes visibility and direct manipulation (Shneiderman & Plaisant, 2010).

7. Improvements in Function Put in place

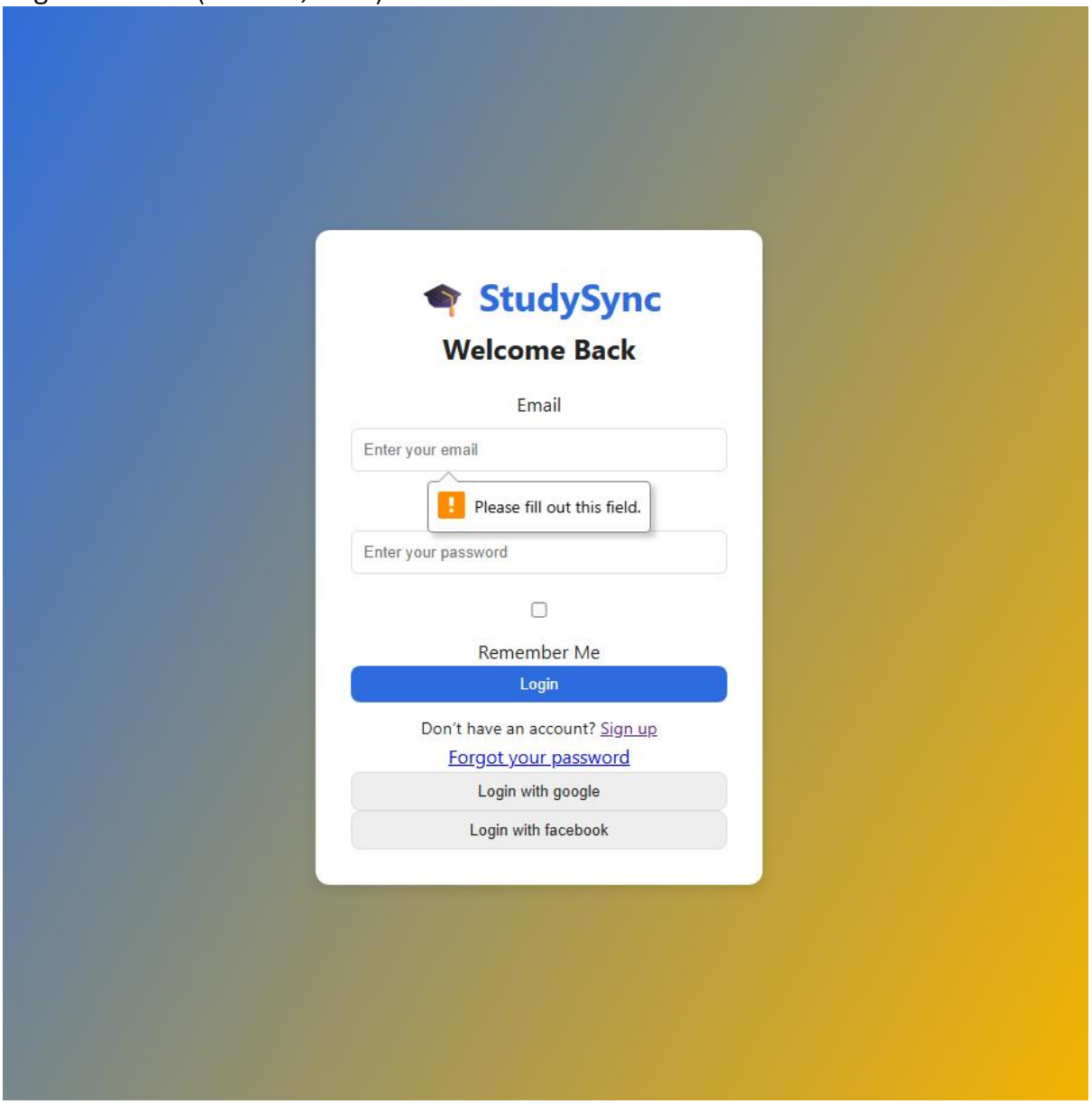
- ✧ **System for signing up and logging in**
  - The authentication model uses localStorage to mimic the creation of an account.
  - This facilitates fundamental user feedback and control (Nielsen, 1994).
- ✧ **Profiles of Students**
  - Students can customize their identities through profile editing, which is a crucial component of user engagement (Shneiderman & Plaisant, 2010).
- ✧ **Management of Tasks**
  - Students can add assignments, note due dates, check their progress, and mark assignments as finished.
- ✧ **Tools for Peer Interaction**
  - Reactions that are liked, commented on, and celebrated enhance community experiences and social engagement.

- ✧ **Tools for Sharing**
  - Includes links to Facebook, X, WhatsApp, and automatically generated group codes.

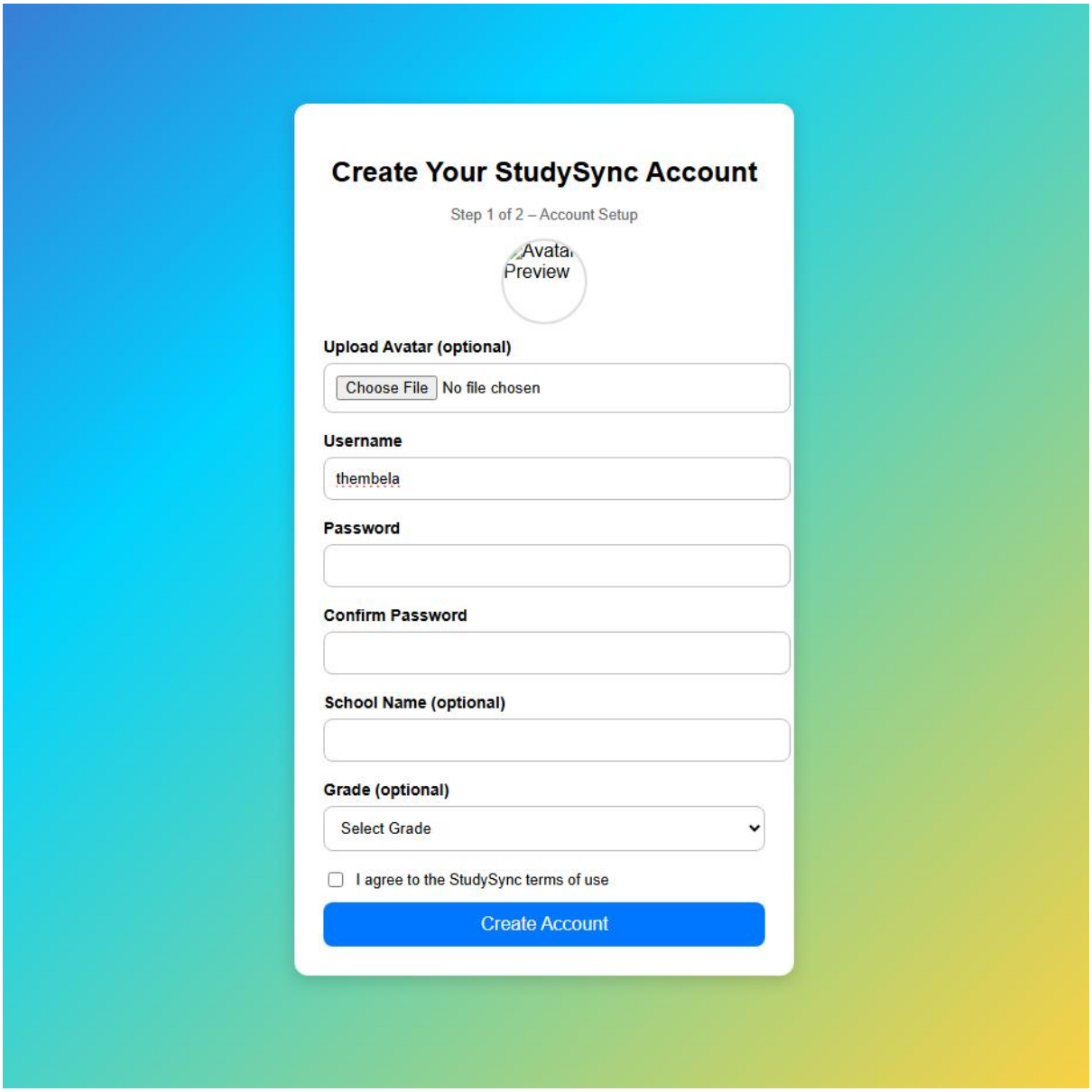
Preview of a Live Video The prototype uses getUserMedia to demonstrate real-time camera access.

8. Screens of the Interface

- ❋ **The Login Page**
  - The straightforward design of the login page promotes learnability and lessens cognitive strain (Nielsen, 1994).



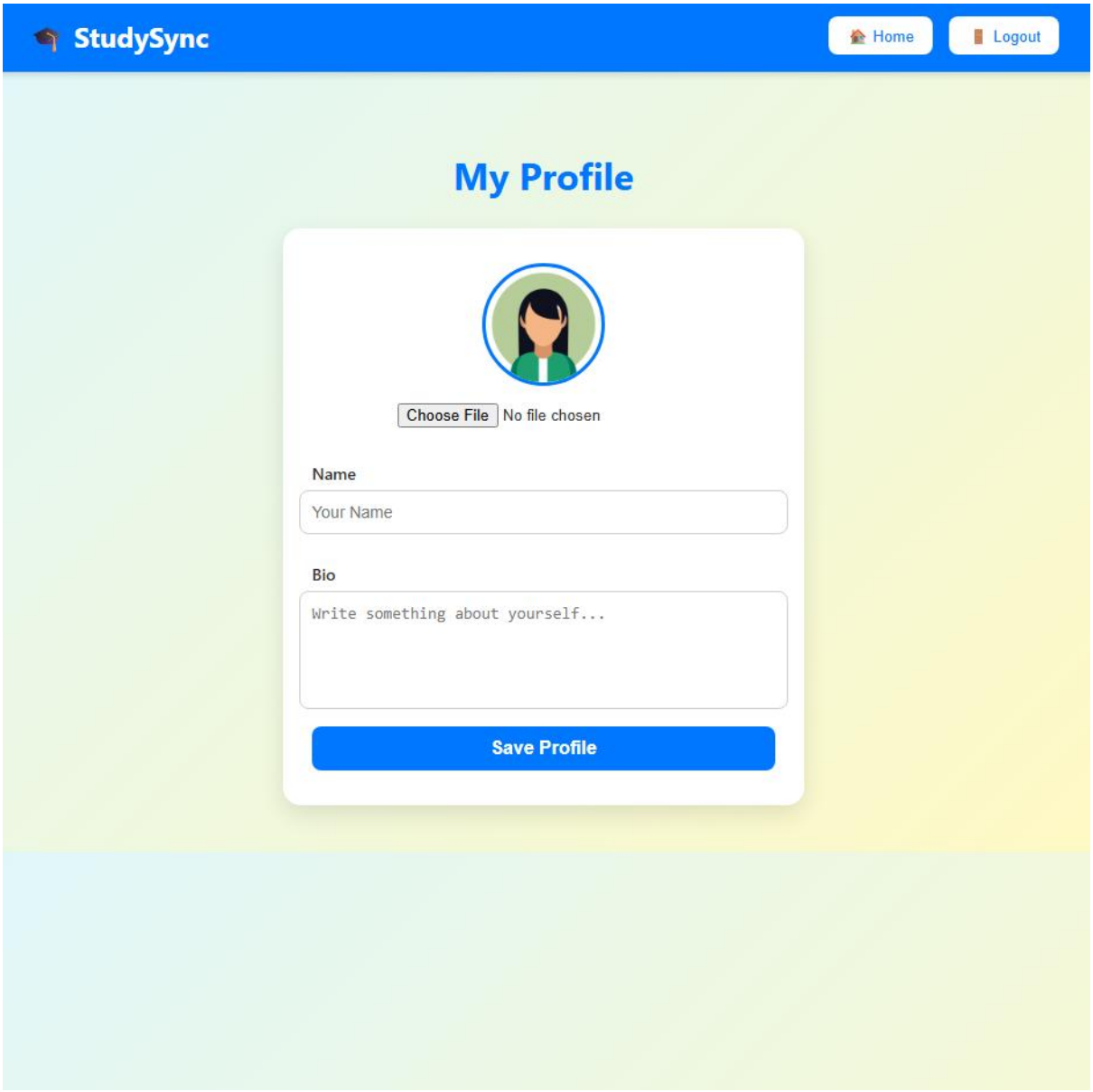
- ❋ **Signup Screen**



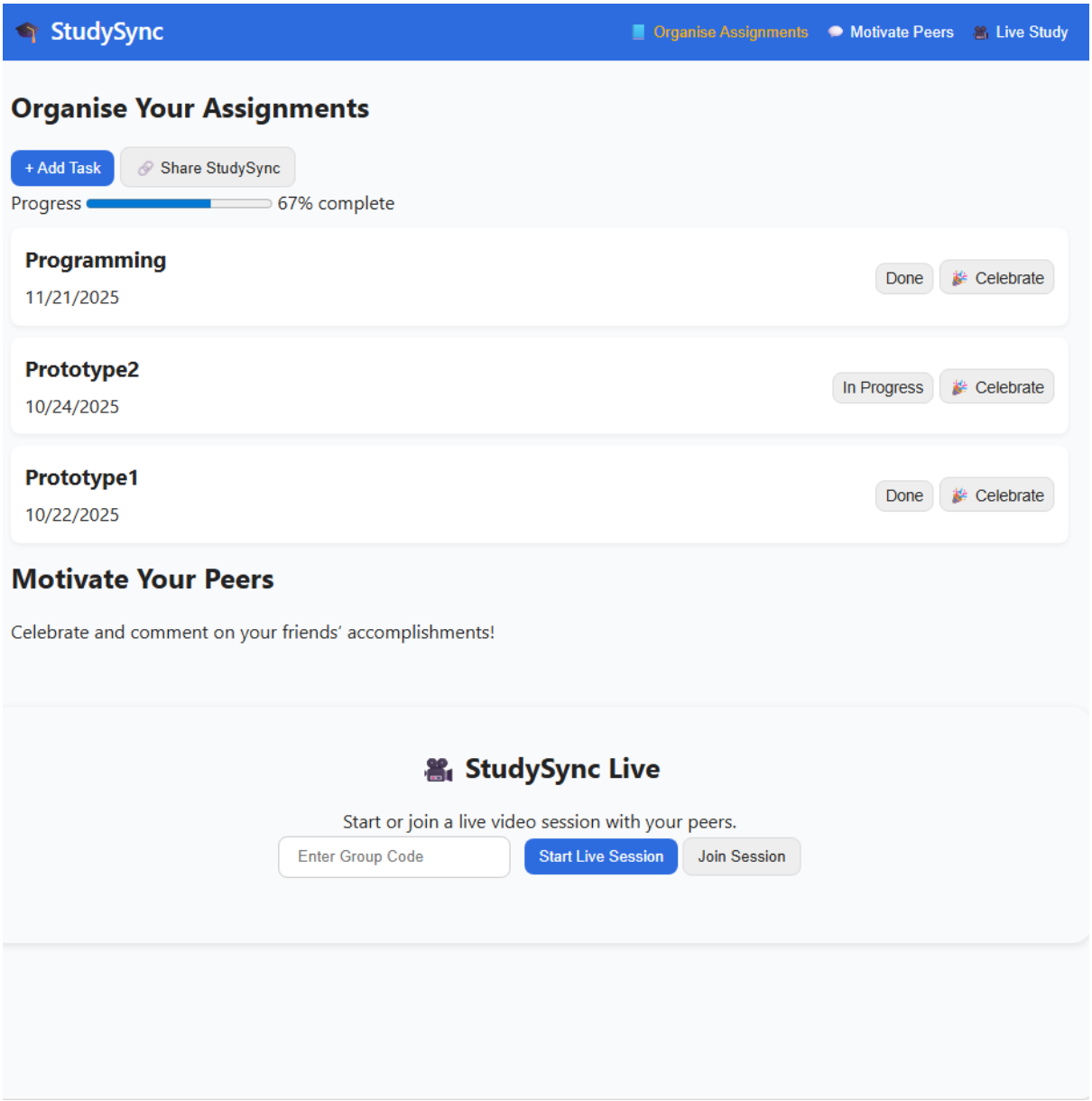
- Clarity and user satisfaction are enhanced by the simple form design (Shneiderman & Plaisant, 2010).

❋ **Profile Page**

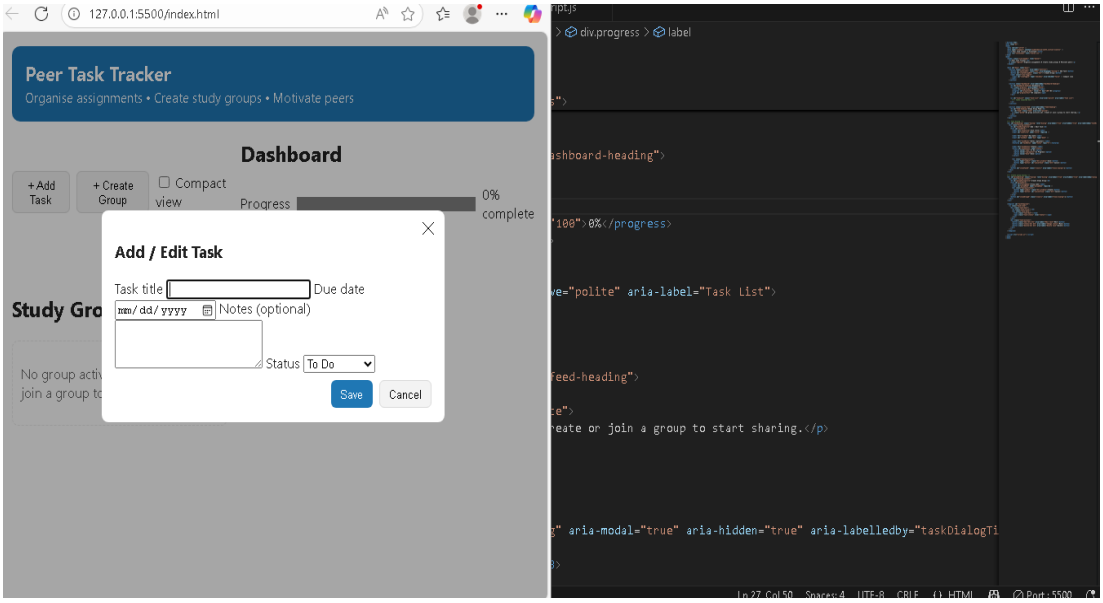
- Features that enhance personalization include the ability for users to update optional school information, alter personal information, and upload avatars.



- **HomeDashboard**
  - Visibility and usability are supported by prominent action buttons and clear navigation (Nielsen, 1994).

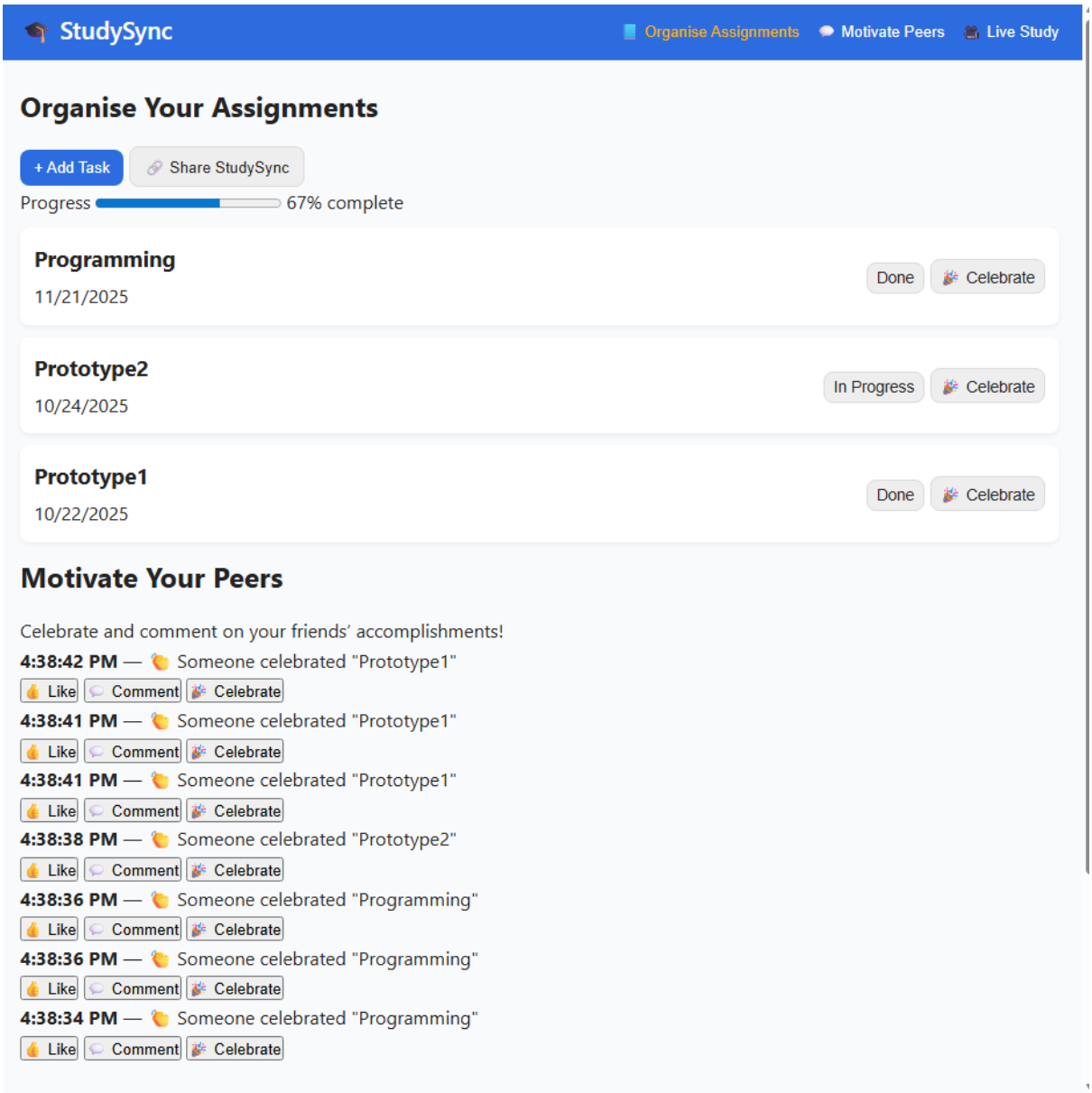


- **Dialogue for Task Creation**
  - By allowing users to create tasks right away, this modal exemplifies direct manipulation (Shneiderman & Plaisant, 2010).



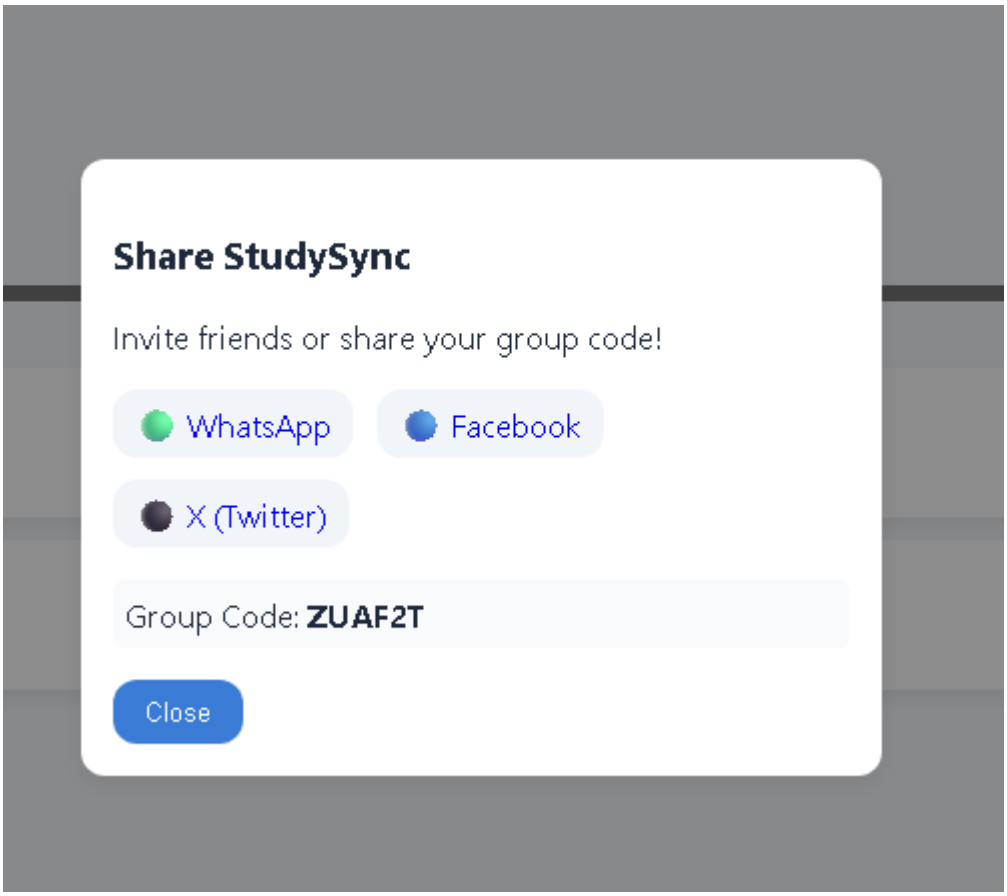
✵ Encourage Peers to Feed

- The feed promotes celebration, emotional connection, communication, and feedback.



❋ Dialogue for Sharing

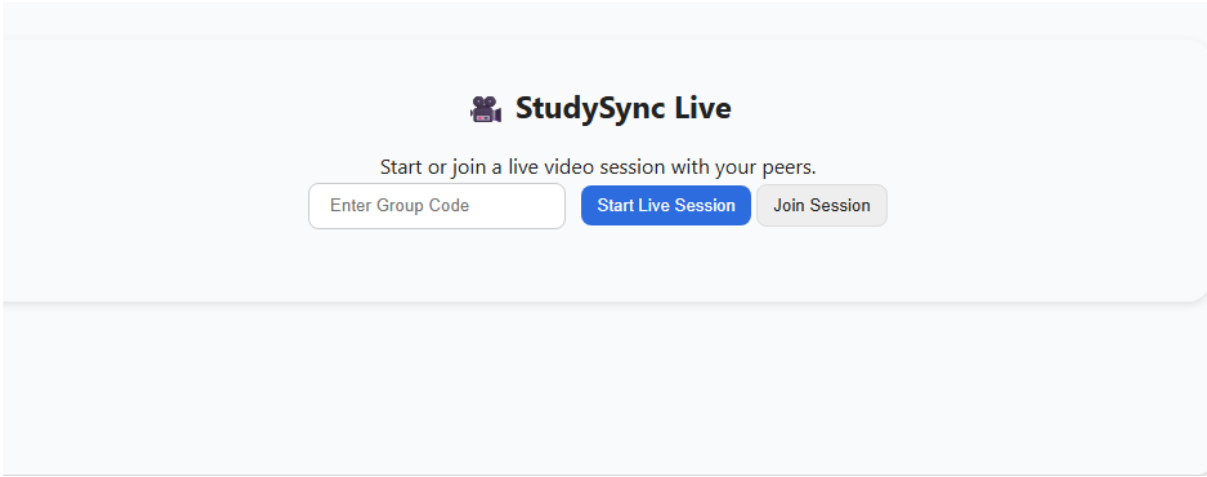
- Links for social sharing adhere to accepted standards for rapid information distribution.



-

❋ **Live Video Prototype Screen**

- Shows responsive interaction feedback and real-time media access.



9. Enhancements to User Experience

Among the enhancements are:

- ✧ Greater color contrast.
- ✧ Greater spacing in the layout.
- ✧ Navigation that is easier to use.
- ✧ Motivated by engagement

These modifications enhance usability and lessen cognitive strain (Nielsen, 1994).

10. Usability Assessment

- ★ **Indicators of Success**
  - • Navigating on your own.
  - • It takes less than three clicks to create a task.
  - • Quick responses and comments.
  - • Instant feedback on a video preview.
- ★ **Problems Found**
  - • HTTPS or localhost are needed for camera preview.
  - • When browser data is deleted, LocalStorage is reset.

Usability inspection techniques suggested by HCI literature were used for assessment (Shneiderman & Plaisant, 2010).

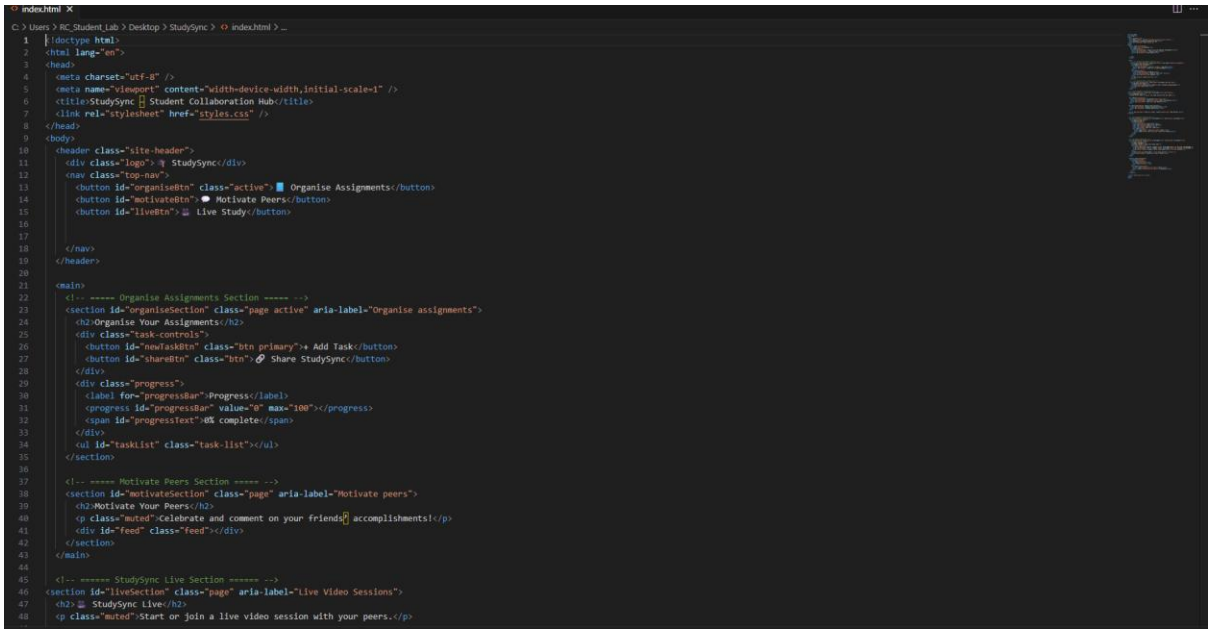
11. WCAG (Accessibility Improvements)

Accessibility features in StudySync include:

- ➢ Alt text for avatars and icons.
- ➢ Keyboard usability.
- ➢ Scalable layouts and readable fonts.
- ➢ Colour matching with high contrast.

These adhere to WCAG 2.0 guidelines (W3C, 2018).

12. Complete Code Interpretation



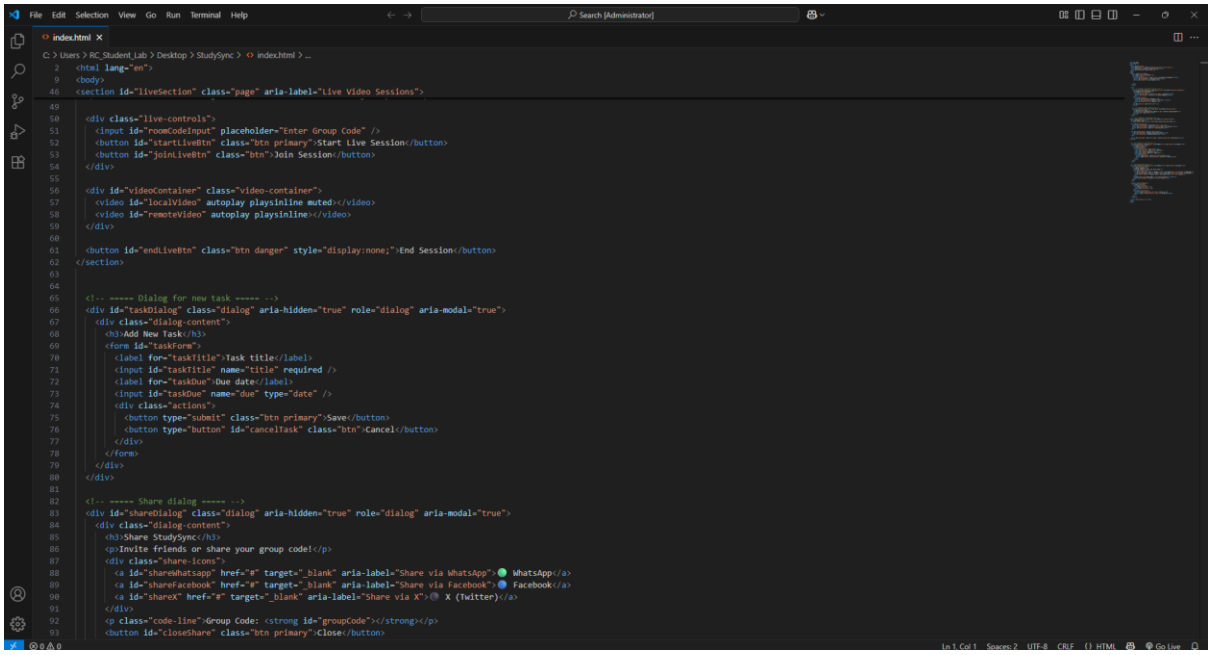
```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>StudySync — Student Collaboration Hub</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="logo">📚 StudySync</div>
    <nav class="top-nav">
      <button id="organiseBtn" class="active">📋 Organise Assignments</button>
      <button id="motivateBtn">💬 Motivate Peers</button>
      <button id="liveBtn">🎥 Live Study</button>


    </nav>
  </header>

  <main>
    <!-- ===== Organise Assignments Section ===== -->
    <section id="organiseSection" class="page active" aria-label="Organise assignments">
      <h2>Organise Your Assignments</h2>
      <div class="task-controls">
        <button id="newTaskBtn" class="btn primary">+ Add Task</button>
        <button id="shareBtn" class="btn">🔗 Share StudySync</button>
      </div>
      <div class="progress">
        <label for="progressBar">Progress</label>
        <progress id="progressBar" value="0" max="100"></progress>
        <span id="progressText">0% complete</span>
      </div>
      <ul id="taskList" class="task-list"></ul>
    </section>

    <!-- ===== Motivate Peers Section ===== -->
    <section id="motivateSection" class="page" aria-label="Motivate peers">
      <h2>Motivate Your Peers</h2>
      <p class="muted">Celebrate and comment on your friends' accomplishments!</p>
      <div id="feed" class="feed"></div>
    </section>
  </main>

  <!-- ====== StudySync Live Section ====== -->
<section id="liveSection" class="page" aria-label="Live Video Sessions">
  <h2>🎥 StudySync Live</h2>
  <p class="muted">Start or join a live video session with your peers.</p>
```



```
<html lang="en">
<body>
<section id="liveSection" class="page" aria-label="Live Video Sessions">

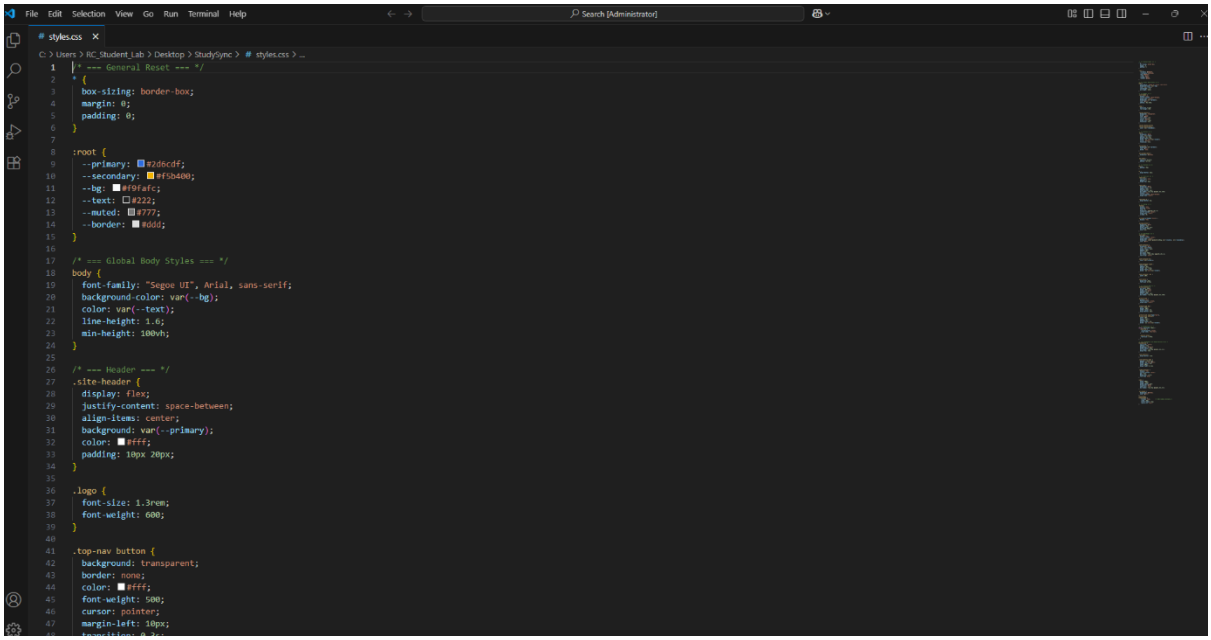
  <div class="live-controls">
    <input id="roomCodeInput" placeholder="Enter Group Code" />
    <button id="startLiveBtn" class="btn primary">Start Live Session</button>
    <button id="joinLiveBtn" class="btn">Join Session</button>
  </div>

  <div id="videoContainer" class="video-container">
    <video id="localVideo" autoplay playsinline muted></video>
    <video id="remoteVideo" autoplay playsinline></video>
  </div>

  <button id="endLiveBtn" class="btn danger" style="display:none;">End Session</button>
</section>


<!-- ===== Dialog for new task ===== -->
<div id="taskDialog" class="dialog" aria-hidden="true" role="dialog" aria-modal="true">
  <div class="dialog-content">
    <h3>Add New Task</h3>
    <form id="taskForm">
      <label for="taskTitle">Task title</label>
      <input id="taskTitle" name="title" required />
      <label for="taskDue">Due date</label>
      <input id="taskDue" name="due" type="date" />
      <div class="actions">
        <button type="submit" class="btn primary">Save</button>
        <button type="button" id="cancelTask" class="btn">Cancel</button>
      </div>
    </form>
  </div>
</div>

<!-- ===== Share dialog ===== -->
<div id="shareDialog" class="dialog" aria-hidden="true" role="dialog" aria-modal="true">
  <div class="dialog-content">
    <h3>Share StudySync</h3>
    <p>Invite friends or share your group code!</p>
    <div class="share-icons">
      <a id="shareWhatsapp" href="#" target="_blank" aria-label="Share via WhatsApp">🟢 WhatsApp</a>
      <a id="shareFacebook" href="#" target="_blank" aria-label="Share via Facebook">🔵 Facebook</a>
      <a id="shareX" href="#" target="_blank" aria-label="Share via X">⚫ X (Twitter)</a>
    </div>
    <p class="code-line">Group Code: <strong id="groupCode"></strong></p>
    <button id="closeShare" class="btn primary">Close</button>
```



```
/* === General Reset === */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

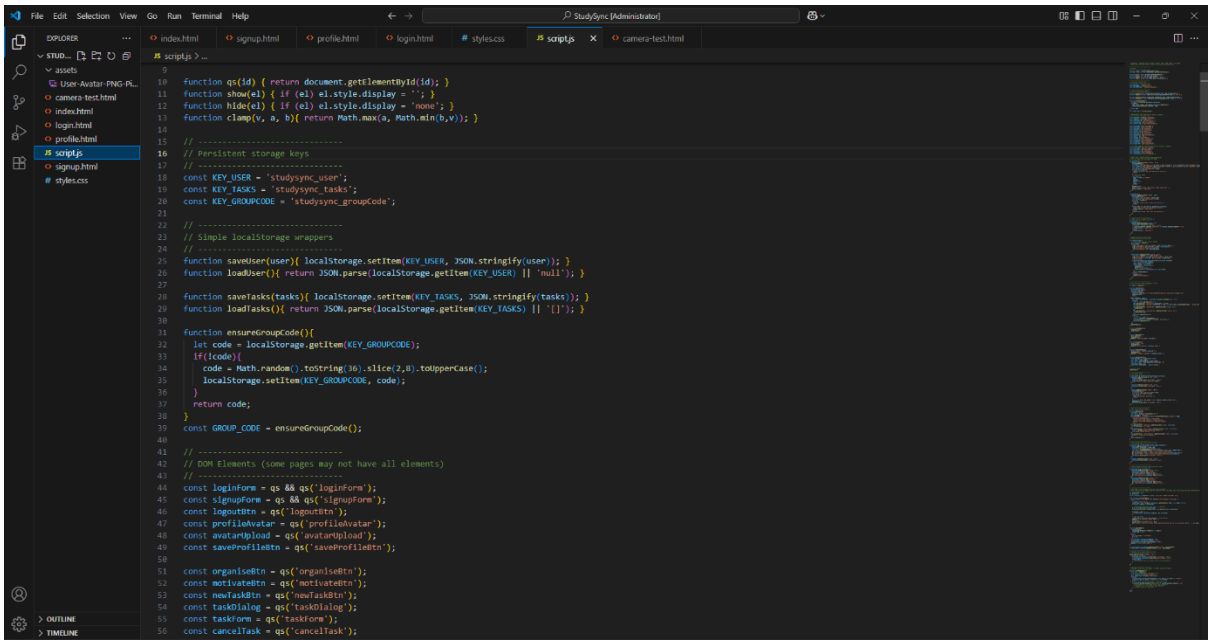
:root {
  --primary: #2d6cdf;
  --secondary: #f5b400;
  --bg: #f0f4fc;
  --text: #222;
  --muted: #777;
  --border: #ddd;
}

/* === Global Body Styles === */
body {
  font-family: "Segoe UI", Arial, sans-serif;
  background-color: var(--bg);
  color: var(--text);
  line-height: 1.6;
  min-height: 100vh;
}

/* === Header === */
.site-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background: var(--primary);
  color: #fff;
  padding: 10px 20px;
}

.logo {
  font-size: 1.3rem;
  font-weight: 600;
}

.top-nav button {
  background: transparent;
  border: none;
  color: #fff;
  font-weight: 500;
  cursor: pointer;
  margin-left: 10px;
```



```
function qs(id) { return document.getElementById(id); }
function show(el) { if (el) el.style.display = ''; }
function hide(el) { if (el) el.style.display = 'none'; }
function clamp(v, a, b){ return Math.max(a, Math.min(b,v)); }

// ------------------------------
// Persistent storage keys
// ------------------------------
const KEY_USER = 'studysync_user';
const KEY_TASKS = 'studysync_tasks';
const KEY_GROUPCODE = 'studysync_groupCode';

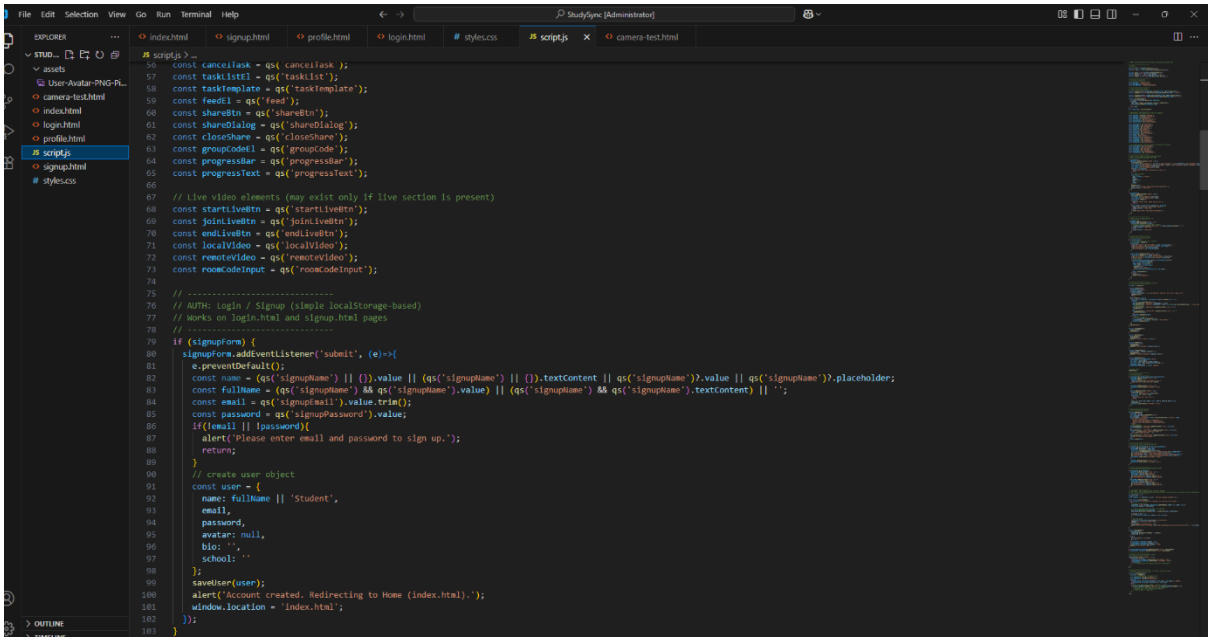
// ------------------------------
// Simple localStorage wrappers
// ------------------------------
function saveUser(user){ localStorage.setItem(KEY_USER, JSON.stringify(user)); }
function loadUser(){ return JSON.parse(localStorage.getItem(KEY_USER) || 'null'); }

function saveTasks(tasks){ localStorage.setItem(KEY_TASKS, JSON.stringify(tasks)); }
function loadTasks(){ return JSON.parse(localStorage.getItem(KEY_TASKS) || '[]'); }

function ensureGroupCode(){
  let code = localStorage.getItem(KEY_GROUPCODE);
  if(!code){
    code = Math.random().toString(36).slice(2,8).toUpperCase();
    localStorage.setItem(KEY_GROUPCODE, code);
  }
  return code;
}
const GROUP_CODE = ensureGroupCode();

// ------------------------------
// DOM Elements (some pages may not have all elements)
// ------------------------------
const loginForm = qs && qs('loginForm');
const signupForm = qs && qs('signupForm');
const logoutBtn = qs('logoutBtn');
const profileAvatar = qs('profileAvatar');
const avatarUpload = qs('avatarUpload');
const saveProfileBtn = qs('saveProfileBtn');

const organiseBtn = qs('organiseBtn');
const motivateBtn = qs('motivateBtn');
const newTaskBtn = qs('newTaskBtn');
const taskDialog = qs('taskDialog');
const taskForm = qs('taskForm');
const cancelTask = qs('cancelTask');
```



```
const cancelTask = qs('cancelTask');
const taskListEl = qs('taskList');
const taskTemplate = qs('taskTemplate');
const feedEl = qs('feed');
const shareBtn = qs('shareBtn');
const shareDialog = qs('shareDialog');
const closeShare = qs('closeShare');
const groupCodeEl = qs('groupCode');
const progressBar = qs('progressBar');
const progressText = qs('progressText');

// Live video elements (may exist only if live section is present)
const startLiveBtn = qs('startLiveBtn');
const joinLiveBtn = qs('joinLiveBtn');
const endLiveBtn = qs('endLiveBtn');
const localVideo = qs('localVideo');
const remoteVideo = qs('remoteVideo');
const roomCodeInput = qs('roomCodeInput');

// ----------------------------
// AUTH: Login / Signup (simple localStorage-based)
// Works on login.html and signup.html pages
// ----------------------------
if (signupForm) {
  signupForm.addEventListener('submit', (e)=>{
    e.preventDefault();
    const name = (qs('signupName') || {}).value || (qs('signupName') || {}).textContent || qs('signupName')?.value || qs('signupName')?.placeholder;
    const fullName = (qs('signupName') && qs('signupName').value) || (qs('signupName') && qs('signupName').textContent) || '';
    const email = qs('signupEmail').value.trim();
    const password = qs('signupPassword').value;
    if(!email || !password){
      alert('Please enter email and password to sign up.');
      return;
    }
    // create user object
    const user = {
      name: fullName || 'Student',
      email,
      password,
      avatar: null,
      bio: '',
      school: ''
    };
    saveUser(user);
    alert('Account created. Redirecting to Home (index.html).');
    window.location = 'index.html';
  });
}
```

- Semantic layout and page structure are provided by **HTML.**
- Visual hierarchy, responsive layouts, themes, and styling are all implemented using **CSS**.
- **JavaScript** oversees all dynamic aspects, such as camera preview, feed replies, task creation, authentication, and notification updates. Recommended UI development strategies are reflected in these design approaches (Shneiderman & Plaisant, 2010).

13. Conclusion

The StudySync Part 3 prototype shows how using HCI principles may produce a useful, accessible, and entertaining platform for student cooperation. The system successfully promotes student enthusiasm and productivity by including accessibility requirements and usability heuristics (Nielsen, 1994; W3C, 2018).

References (Harvard Style)

Nielsen, J. (1994). *Usability Engineering*. San Diego: Academic Press.
Shneiderman, B. & Plaisant, C. (2010). *Designing the User Interface: Strategies for Effective Human–Computer Interaction*. Boston: Pearson.
World Wide Web Consortium (W3C). (2018). *Web Content Accessibility Guidelines (WCAG) 2.0*. Available at: https://www.w3.org/TR/WCAG20/